6 **(a)** **(i)** Explain what is meant by the term *nuclear binding energy*.

.....

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.....

.....

.....[2]

(ii) A nucleus of strontium-90 has a mass of 89.907 *u*. It contains 38 protons. The mass of a neutron is 1.00867 *u* and the mass of a proton is 1.00783 *u*.

Calculate the nuclear binding energy per nucleon of strontium-90.

binding energy per nucleon = ... J [2]

(b) The nuclear fission of uranium-235 can be described by

$$^{1}_{0}n + {}^{235}_{92}U \longrightarrow {}^{144}_{56}Ba + {}^{89}_{36}Kr + 3{}^{1}_{0}n.$$

Table 6.1 shows the masses of the particles involved in this reaction.

Table 6.1

particle	mass/u
$^{1}_{0}n$	1.008665
$^{89}_{36}Kr$	88.917636
$^{144}_{56}Ba$	143.922953
$^{235}_{92}U$	235.043930

Determine the energy released by the fission of 15.0 kg of uranium-235.

energy = ... J [4]

[Total: 8]